

****Inventory Management and Supply Chain Optimization****

****Literature Review****

****Tang (2024)**** conducted a pivotal study that directly addresses the core challenge of demand forecasting in inventory management. The author proposed a hybrid model combining ****LightGBM**** and ****PSO-LSTM****, leveraging big data analytics to enhance forecast accuracy. This framework was integrated into a two-level multi-cycle inventory control model, leading to significant empirical improvements for Company H, including an increase in inventory turnover from 8.2 to 9.2 and an On-Time In-Full (OTIF) rate from 97.9% to 99.3%. This work demonstrates the tangible operational benefits of combining gradient boosting with metaheuristic-optimized deep learning for supply chain optimization. ***(2024)***

****Park, Turner, and Becker (2025)**** shifted the paradigm from predictive to prescriptive analytics by formulating inventory management as a ****Markov Decision Process (MDP)****. Their ****Deep Reinforcement Learning (DRL)**** agent autonomously learned replenishment policies that dynamically balanced holding costs, shortage costs, and service levels, outperforming traditional methods like EOQ in volatile retail environments. This approach highlights the potential for autonomous, self-optimizing inventory systems. ***(2025)***

****Khedr and Rani (2024)**** provided a comprehensive review of ****ML and DL applications**** across the entire supply chain, from supplier selection to transportation. They confirmed the superiority of models like Random Forests and Neural Networks in improving forecasting and logistics but also critically highlighted persistent barriers such as model interpretability, data quality issues, and the computational complexity of deploying these solutions at scale. ***(2024)***

****Namira et al. (2022)**** developed a hybrid decision support system that integrated ****ARIMA, NARNN, and XGBoost**** for price and demand forecasting, coupled with a combinatorial optimization module for strategic inventory decisions. Their framework proved effective in maximizing profitability while mitigating the risks of overstocking and stockouts, showcasing the power of hybridizing statistical and machine learning techniques. ***(2022)***

****Sayyad et al. (2024)**** focused on the e-commerce domain, which is characterized by high-dimensional categorical data. They employed the ****CatBoost**** algorithm to effectively handle features like product categories and customer segments, achieving superior performance in predicting demand surges and optimizing logistics compared to traditional models, thus illustrating the importance of selecting algorithms suited to specific data structures. ***(2024)***

****Ait Ben Hamou et al. (2024)**** demonstrated the application of ML in production scheduling, a key upstream component of the supply chain. Using ****XGBoost**** and other models, their decision support system accurately predicted production costs and optimized scheduling across non-identical parallel machines, achieving a high R^2 value of 0.96 and improving overall manufacturing throughput. ***(2024)***

****Onasanya et al. (2024)**** proved that the benefits of ML are accessible to ****Small and Medium Enterprises (SMEs)****. By utilizing ****Random Forest and Support Vector Regression (SVR)****, their system provided accurate demand forecasts and optimal reorder points, enabling SMEs to achieve cost savings and service level improvements without requiring extensive computational resources. ***(2024)***

****An anonymous systematic review (2024)**** cataloged the expanding role of ****Artificial Intelligence (AI)****, including ****NLP and Computer Vision****, in supply chain management. The review underscored AI's impact on sustainability, risk mitigation, and logistics, while

simultaneously calling for greater emphasis on **Explainable AI (XAI)** and ethical frameworks to ensure responsible deployment. **(2024)**

Collectively, this body of literature establishes that machine learning and big data analytics are fundamentally transforming supply chain management. The trend is moving from single-model forecasting towards sophisticated hybrid frameworks and autonomous, reinforcement learning-based systems that offer not only improved predictions but also prescriptive, real-time decision support for end-to-end optimization.

References

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